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LBA3 Predictors of benefit from local consolidative therapy and patterns of failure for metastatic EGFR-mutant NSCLC: A secondary analysis of NorthStar, a phase II randomized clinical trial

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Background: The NorthStar trial demonstrated improved progression free survival (PFS) with local consolidative therapy (LCT; radiotherapy [RT] or surgery) added to Osimertinib (Osi) in metastatic EGFR-mutant NSCLC (25.4 vs 17.5 months [mo]; HR 0.66, 90% CI 0.50−0.87; p=0.025). Evaluation of RT techniques, patterns of failure, and predictors of LCT benefit are of immense interest for patient (pt) selection.

Methods: This secondary analysis assessed RT dosimetry, outcomes, and failure patterns after LCT. Cox proportional-hazards modeling evaluated factors associated with PFS.

Results: Among 119 pts without progression after 6-12 weeks of Osi, 63 were randomized to Osi alone and 56 to Osi + LCT. LCT modalities were RT alone, 33 (59%); RT + surgery, 6 (11%); surgery alone, 17 (30%). 76% of pts continued Osi during RT. Common RT regimens were 52.5 Gy/15 fx, 50 Gy/4 fx, 60 Gy/15 fx, and 45 Gy/15 fx. Pts without residual thoracic nodal involvement on post-induction diagnostic CT scan derived the greatest benefit, with median PFS of 41.5 mo with Osi + LCT versus 19.6 mo with Osi alone (HR 0.43, 90% CI 0.23−0.78; p=0.008). Similarly, among patients without persistent pleural effusion on post-induction diagnostic CT scan, median PFS was 32.7 versus 22.3 mo (HR 0.63, 90% CI 0.39−1.02; p=0.057). In contrast, no PFS benefit was observed in pts with persistent thoracic nodal disease (19.0 vs 17.5 months; HR 0.94, 90% CI 0.61−1.46; p=0.41) or persistent pleural effusion (15.3 vs 12.9 months; HR 0.90, 90% CI 0.52−1.55; p=0.37) after induction Osi. Among 37 recurrences after LCT, 7 (19%) were locoregional, 23 (62%) were distant, and 7 (19%) were both; 23 (62%) occurred at new metastatic sites. Among the subset who had RT, there were 24 recurrences, only 2 (8%) were in-field, whereas 22 (92%) were out-of-field failures.

Conclusions: Addition of LCT to Osi significantly prolonged PFS compared with Osi alone. Clearance of thoracic nodal disease and pleural effusion following induction Osi emerged as potential predictors of benefit from LCT. Failures after LCT were predominantly distant, occurring outside the RT field and most commonly at new metastatic sites.

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LBA4 Sacituzumab tirumotecan (sac-TMT) in patients (pts) with previously treated advanced EGFR-mutated (EGFRm) non-small cell lung cancer (NSCLC): Final overall survival (OS) analysis from the randomized OptiTROP-Lung03 study

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Background: Sac-TMT is a trophoblast cell surface antigen 2 (TROP2)-targeting antibody-drug conjugate (ADC) with a unique bifunctional linker. The OptiTROP-Lung03 study (NCT05631262) previously demonstrated statistically significant progression-free survival (PFS) and OS benefits with sac-TMT vs docetaxel in previously treated EGFRm NSCLC (hazard ratio [HR] for PFS: 0.30, 95% confidence interval [CI] 0.20 to 0.46, one-sided p<0.001; HR for OS: 0.49, 95% CI 0.27 to 0.88, one-sided p=0.007; Fang *et al*., BMJ 2025). Here, we report the results of pre-specified final OS analysis from this study, including updated PFS and safety.

Methods: Pts with advanced EGFRm NSCLC who had progressed after EGFR tyrosine kinase inhibitor (TKI) and platinum-based chemotherapy were randomized (2:1) to receive sac-TMT 5 mg/kg every 2 weeks or docetaxel 75 mg/m^2 every 3 weeks. Pts with verified progression on docetaxel could be crossed over to receive sac-TMT if eligible. The crossover-adjusted OS was derived using the rank-preserving structural failure time (RPSFT) model.

Results: A total of 137 pts were randomized to receive sac-TMT (n=91) or docetaxel (n=46). As of December 11, 2025, with a median follow-up of 23.8 months (mo), median PFS assessed by investigator was 7.9 mo vs 2.8 mo (sac-TMT vs docetaxel; HR 0.23, 95% CI 0.15, 0.35) and median OS was 20.0 mo vs 13.5 mo (HR 0.63, 95% CI 0.40, 0.98). In the docetaxel group, 41.3% of pts received sac-TMT as subsequent anti-cancer therapy. The adjusted median OS by the RPSFT model was 20.0 mo vs 11.2 mo (HR 0.45, 95% CI 0.28, 0.73; 18-mo OS rate: 54.7% vs 9.1%). Grade $\geq$ 3 treatment-related adverse events occurred in 60.4% vs 73.9%, and treatment-related serious adverse events in 20.9% vs 41.3% of pts (sac-TMT vs docetaxel) respectively. No cases of interstitial lung disease were reported in the sac-TMT group.

Conclusions: Sac-TMT continues to demonstrate clinically meaningful and significant improvements in PFS and OS compared to docetaxel, with a manageable safety profile in pts with previously treated advanced EGFRm NSCLC. These results underscore sac-TMT as a promising new treatment option for this population.

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